

Week 6 – Graded Project Report

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Module: Large Language Models & Embeddings

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1. Introduction

This project explores the use of Large Language Models (LLMs) and word embeddings through practical experimentation using Hugging Face and Gensim. The objective was to understand how language models generate text, how tokens are processed, how prompts influence output, and how semantic similarity can be computed using vector representations. The project demonstrates how modern AI models process language and extract meaning using probability-based generation and geometric embeddings.

2. Section A – LLM Foundations & Hugging Face

◆ Text Generation (DistilGPT-2 / Qwen)

Output 1: AI is transforming industries by bringing companies to the table over a number of different technologies and technologies.

Output 2: AI is transforming industries by harnessing innovation for what we call the consumer, who also have tremendous potential for growth and economic growth.

Output 3: AI is transforming industries by adopting a more traditional approach for the management of consumer and retail businesses. Such enterprises also invest heavily in their automation systems, for example adding new features of automation automation systems such as

What I Learned:

- LLMs generate text using statistical probability, not factual reasoning.
 - Each generation varies because sampling introduces randomness.
 - Parameters such as temperature and max length control creativity and output size.
-

◆ Tokenisation

🔍 Collapse Output ^

```
Tokens: ['LL', 'Ms', 'Ġare', 'Ġpowerful', 'Ġtools', 'Ġfor', 'Ġnatural', 'Ġlanguage', 'Ġunderstanding', '.']  
Token IDs: [3069, 10128, 389, 3665, 4899, 329, 3288, 3303, 4547, 13]  
Sequence Length: 10
```

What I Learned:

- Sentences are broken into *tokens* rather than full words.
 - Special markers like Ġ represent spaces.
 - Each token is mapped to a unique integer ID used during model computation.
-

3. Section B – Prompt Engineering

◆ Tasks Performed:

- Summarisation ($\leq$ 30 words)
- Question Answering
- Creative 4-line poem

USING DISTRICTGPT

PROMPT: Strictly answer in 30 words: Artificial intelligence is changing healthcare, How?
Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
RESPONSE: Strictly answer in 30 words: Artificial intelligence is changing healthcare, How?

Now the main problem I hear is the number of "hints" that are not really coming to the United States as of the end of this year. Here are some of the things you should think about when trying to take your healthcare situation to the next level in your life:

Health Care – When a healthcare provider gets the wrong information or the right type of healthcare, then they can be just a little too big or too slow.

You may be thinking that healthcare is growing, because they need more providers right now, and they

PROMPT: Strictly answer in 30 words: Who invented Python and in which year?
Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.
RESPONSE: Strictly answer in 30 words: Who invented Python and in which year? What have you been doing for a quarter of the way through the first half of the past year? Please help us get ready.

A few days ago I got into Google Glass again and found a great and awesome website, which [Ill](#) try to try my best to help support. It was a huge undertaking. The site includes very useful tools for navigating the world, with a number of great projects, with a small library (you could use one): this is one of my projects that has helped many of you and many others that will help me out quite a bit. And it is really not a question that is the answer to any question, and the answer is no surprise and no one should

PROMPT: Strictly answer in 30 words: Write a 4-line poem about AI and humanity.
RESPONSE: Strictly answer in 30 words: Write a 4-line poem about AI and humanity. I'm talking about some of the crazily awful things that happen to our children, from our little boys to our parents to our grandmothers. And they will have to spend their entire childhood playing. It might be a little silly to just say that you're so cruel. But it's going to be interesting to see how things happen to humanity.

USING QWEN

```
=====
TASK: Summarisation
-----
```

```
AI enhances diagnostics, personalizes treatment, streamlines
operations, and improves drug discovery through data analysis,
predictive modeling, and automation, increasing efficiency, accuracy,
and access to care.
```

```
=====
TASK: Q&A
-----
```

```
Guido van Rossum invented Python in 1991.
```

```
=====
TASK: Creative Poem
-----
```

```
AI mirrors minds, learns from human dreams,
Yet lacks soul, emotion, or true sense.
We guide its thoughts, it serves our needs—
Together, we shape tomorrow's chance.
```

What I Learned:

- Prompt clarity strongly affects output quality.
- Structured prompts produce better results.
- Instruction-style phrasing improves accuracy.
- Creative wording leads to diverse and imaginative outputs.

4. Section C – Embeddings with Gensim

◆ Word Embeddings (GloVe)

```
WORD: king
First 10 vector values: [ 0.50451  0.68607 -0.59517 -0.022801  0.60046 -0.13498 -0.08813
 0.47377 -0.61798 -0.31012 ]
Top 5 similar words:
('prince', 0.8236179351806641)
('queen', 0.7839043140411377)
('ii', 0.7746230363845825)
('emperor', 0.7736247777938843)
('son', 0.766719400882721)

WORD: queen
First 10 vector values: [ 0.37854  1.8233 -1.2648 -0.1043  0.35829  0.60029 -0.17538
 0.83767 -0.056798 -0.75795 ]
Top 5 similar words:
('princess', 0.8515166640281677)
('lady', 0.805060863494873)
('elizabeth', 0.787304162979126)
('king', 0.7839043140411377)
('prince', 0.7821861505508423)

WORD: diamond
First 10 vector values: [-0.4958  0.78421 -0.606  1.3967  0.28888 -0.2058 -0.10745 -0.33252
 1.3608  0.15091]
Top 5 similar words:
('gold', 0.7714747190475464)
('diamonds', 0.7662666440010071)
('gem', 0.7375268936157227)
('silver', 0.72101891040802)
('jewel', 0.7102220058441162)
```

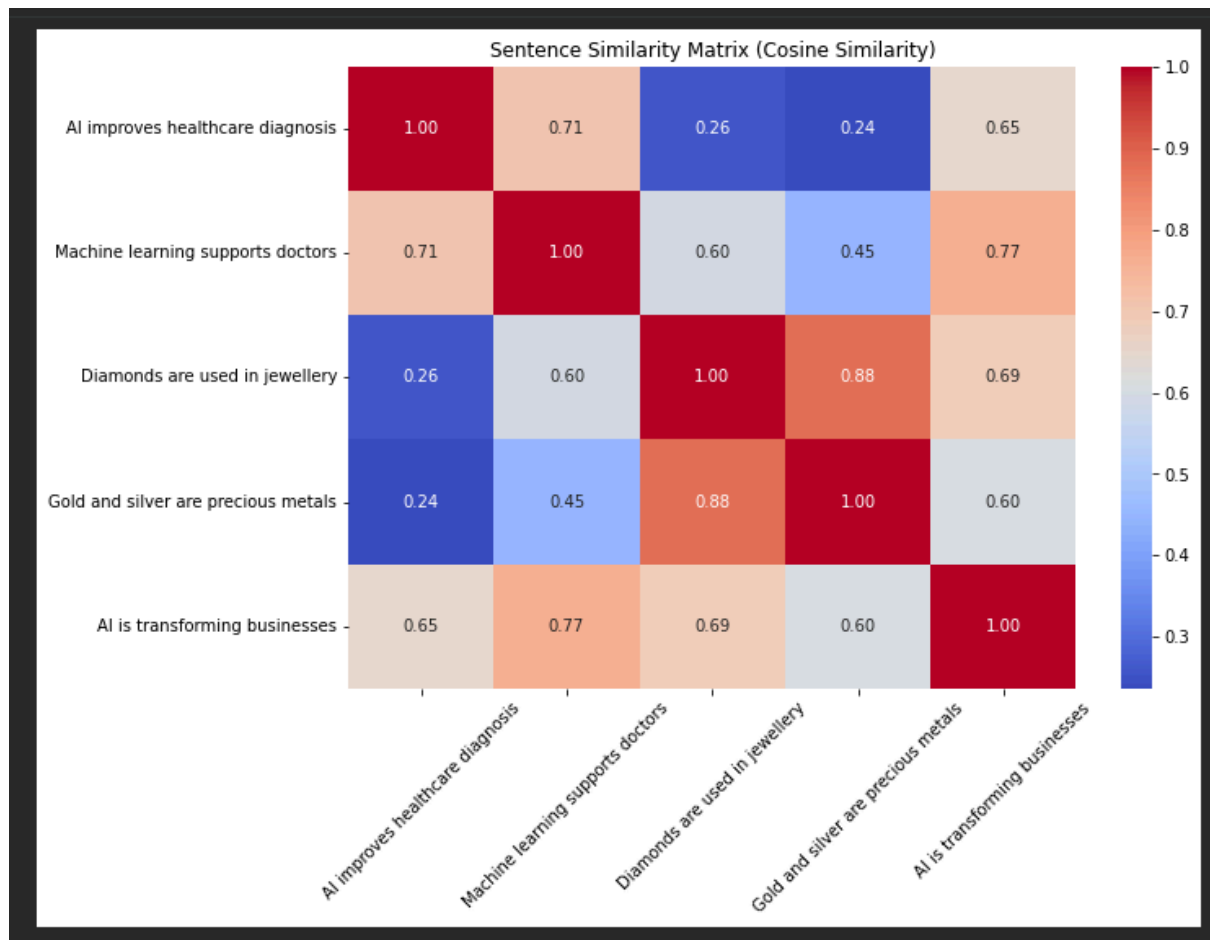
What I Learned:

- Words are represented as numerical vectors.
- Similar words are closer in vector space.
- Semantic relationships such as *king–queen* are mathematically encoded.

◆ Sentence Embeddings & Similarity Matrix

Screenshot Required:

```
[[1.          0.7093215  0.25594014 0.23688862 0.64893836]
 [0.7093215  0.99999994 0.5972236  0.45121273 0.7659094 ]
 [0.25594014 0.5972236  1.          0.8809316  0.6871816 ]
 [0.23688862 0.45121273 0.8809316  0.99999994 0.6039631 ]
 [0.64893836 0.7659094  0.6871816  0.6039631  1.0000001 ]]
```



What I Learned:

- Sentence vectors can be formed by averaging word vectors.
- Cosine similarity shows how related two sentences are.
- Topic-related sentences have higher similarity scores.

5. Section D – Application Demonstrations

◆ Sentiment Analysis

```
Input text:
The new product launch was a success, but some customers complained about the slow delivery time.

Sentiment result:
[{'label': 'NEGATIVE', 'score': 0.9982717037200928}]
```

◆ Summarisation

Original text:

Our company has been investing heavily in artificial intelligence to automate customer support, optimise supply chains, and personalise marketing campaigns. These initiatives have reduced operational costs, improved response times, and increased overall customer satisfaction across multiple regions.

Summary:

Our company has been investing heavily in artificial intelligence to automate customer support, optimise supply chains, and personalise marketing campaigns. These initiatives have reduced operational costs, improved response times,

◆ Translation

```
Original (EN):
Artificial intelligence is helping businesses make faster and more accurate decisions.

Translated (FR):
L'intelligence artificielle aide les entreprises à prendre des décisions plus rapides et plus précises.
```

What I Learned:

- Transformers can classify emotions in reviews.
 - Large texts can be condensed automatically.
 - Multiple languages can be translated in real time.
-

6. Key Observations: LLMs vs Embeddings

LLMs	Embeddings
Generate text	Measure similarity
Non-deterministic	Deterministic vectors
Context-based	Geometry-based
Predict next token	Compare meaning

Summary:

LLMs are generative systems that predict language.

Embeddings are analytic systems that represent meaning geometrically.

7. Real-World Applications of Embeddings & Vector Search

Use Cases:

- Recommendation Systems
- Search Engines
- Chatbots
- Document Matching
- Fraud Detection
- Customer Review Clustering

Business Value:

Vector embeddings allow companies to:

- Detect similar documents.
 - Identify related products.
 - Match questions to answers faster.
 - Implement intelligent search systems.
-

8. Conclusion

This project demonstrated how LLMs generate human-like language and how embeddings mathematically represent meaning. Together, these technologies enable automation, decision support, and intelligent systems that transform industries such as finance, healthcare, customer service, and e-commerce.